

Airline Tweet Sentiment Analysis

End-to-End Text Analytics
Pipeline · Twitter US
Airline Sentiment Corpus



Agenda

01	Business Framing, Data & Text Representation
02	Supervised Classification (Lasso, XGBoost, Threshold Analysis)
03	Unsupervised Analysis – LDA Topics & Clustering
04	Prompt Engineering Experiment
05	Conclusion: Bias & Ethical Reflection

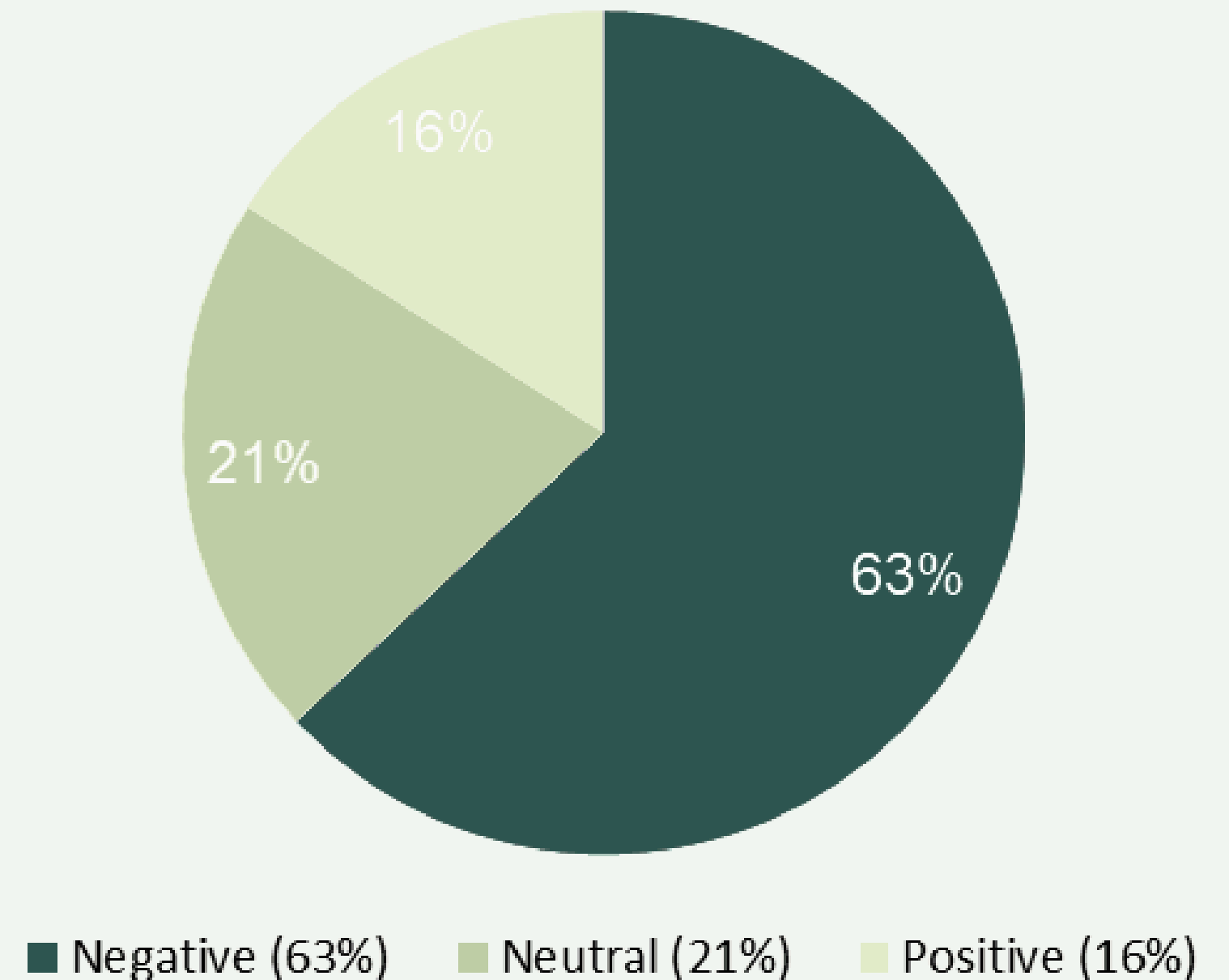
Business framing & Data

Business Question: Can we automatically classify whether a tweet about our airline is negative, so the customer-service team can prioritize outreach?



Target Variable:

- sentiment_binary: 1 = negative, 0 = non-negative
- Derived from airline_sentiment column
- Business priority: Maximize recall on the negative class



14,640

Total Tweets

6

Airlines

Feb 2015

Collection date

0 Missing

in text column

Text processing pipeline ↗

Token counts

Before stop-word removal: 12,070

After stop-word removal: 11,432

After letters/length filter: 8,926



Text representation

BOW VS TF-IDF?

Bag of Words (BOW)

- Counts raw word frequency per document
- High-frequency words (e.g. 'flight') dominate
- 14,124 docs $\times$ 8,926 terms | Sparsity: 99.9%
- Cannot distinguish tweet-specific signal

TF-IDF ✓ Chosen

- Down-weights common terms across docs
- Up-weights distinctive words per document
- Same dimensions , only values differ
- Better signal for rare complaint words

TF-IDF was chosen because it gives more weight to words that actually signal a complaint, like "stranded", "horrible", and "cancelled", rather than words like "flight" that show up in every tweet regardless of how the person feels.

Majority-Class Baseline & Lasso Logistic Regression

Majority-Class Baseline

64.5%

Accuracy

1.000

Recall

0.784

F1

⚠ Always predicts 'negative' — sets the performance floor

Lasso ($\lambda = 0.01$) — Chosen Model

Lambda	Precision	Recall	F1
0.001	N/A	0.000	N/A
0.010 ✓	0.698	0.975	0.814
0.050	0.645	1.000	0.784

Top Negative Predictors

cancelled

hours

worst

hold

delayed

rude

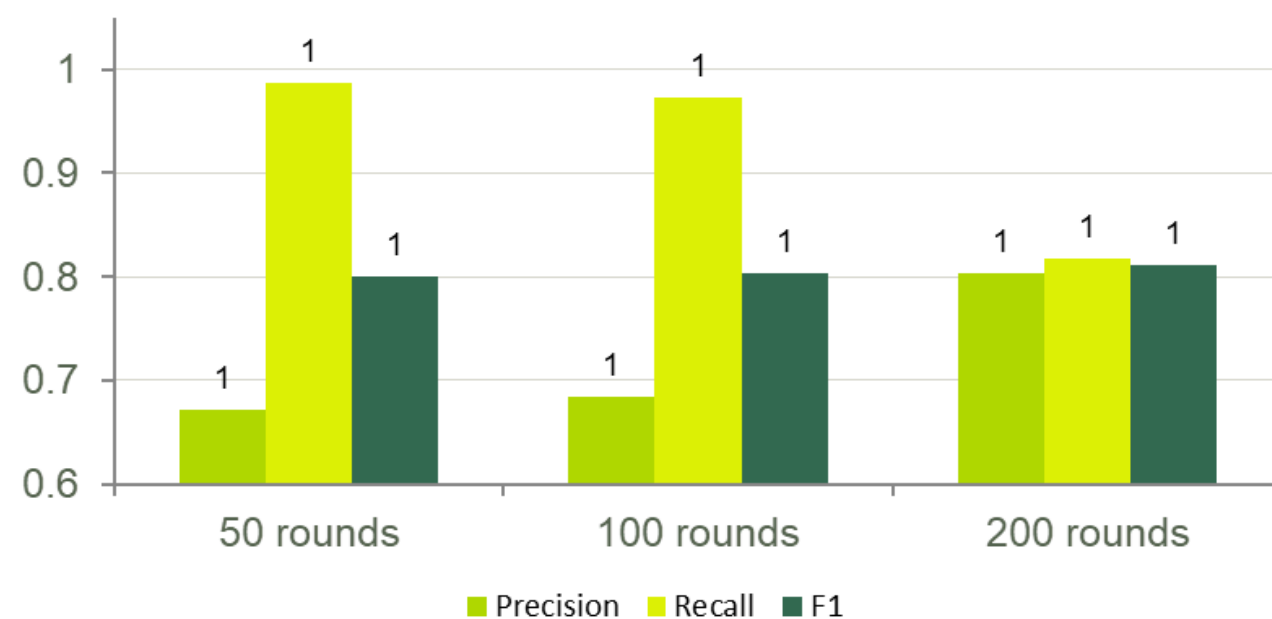
stuck

Model B

XGBoost ↗

- Chosen over Random Forest: faster on sparse text matrices, strong probability calibration for threshold tuning
- Tuned nrounds $\in \{50, 100, 200\}$ with eta = 0.1, max_depth = 6
- Selected nrounds = 100, nearly identical F1 to 200, lower overfitting risk, faster inference

Hyperparameter Tuning: nrounds



Chosen Model Results (nrounds = 100)

0.685

Precision

0.804

F1

0.973

Recall

0.694

Accuracy

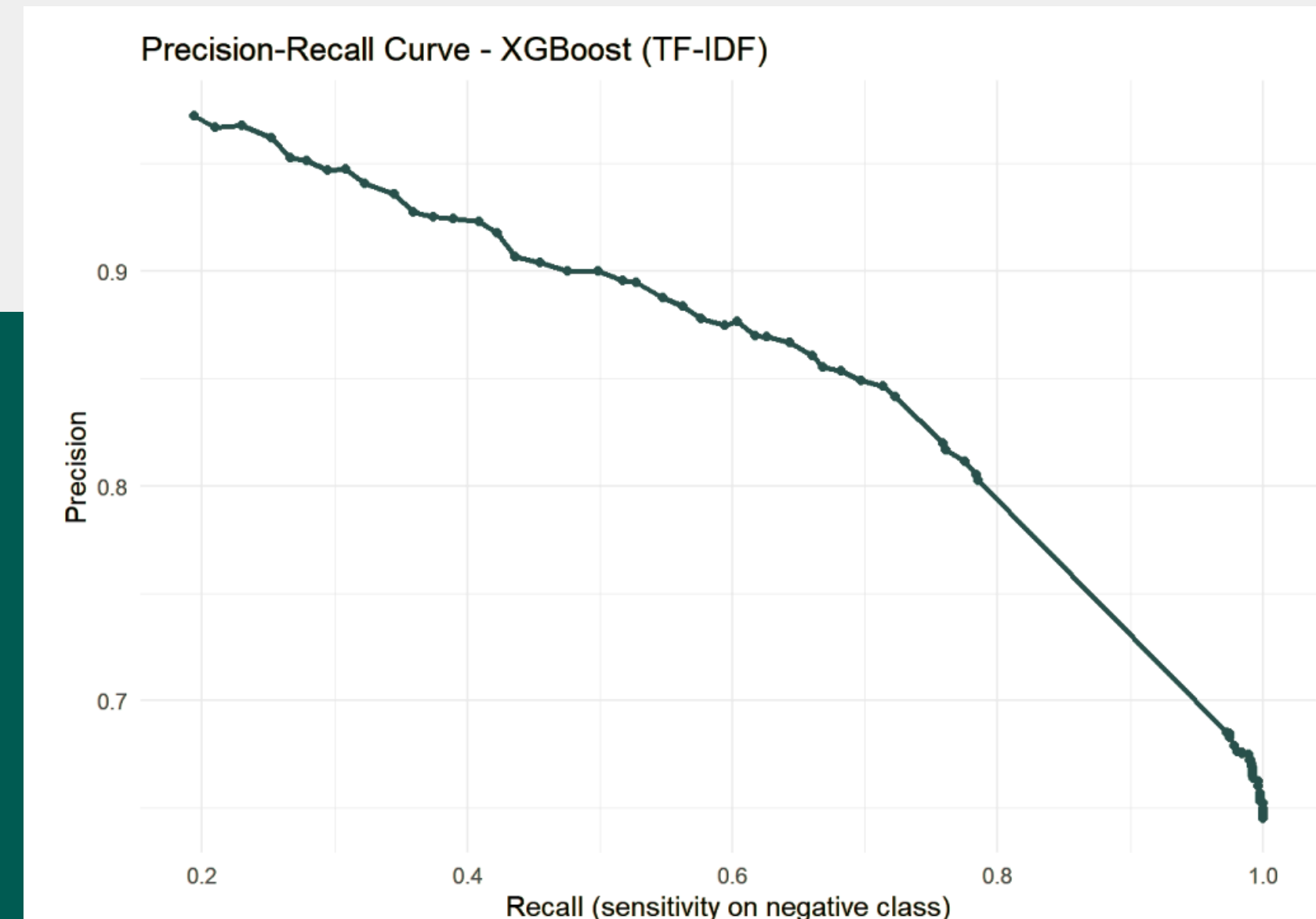
Representation Experiment & Threshold Analysis

BOW vs TF-IDF (XGBoost):

Representation	Precision	Recall	F1
TF-IDF ✓	0.685	0.973	0.804
BOW	0.777	0.825	0.8

Threshold Analysis (XGBoost TF-IDF)

Threshold	Precision	Recall	F1
0.30 ✓ Recommended	0.662	0.997	0.796
0.5	0.685	0.973	0.804



At threshold = 0.30: catches 99.7% of negative tweets. A false positive wastes seconds; a missed complaint may go viral.

Model Comparison & Error Analysis

Model	Precision	Recall	F1	Key Takeaway
Majority-Class Baseline	0.645	1	0.784	Performance floor; F1 inflated by imbalance
Lasso ($\lambda=0.01$)	0.698	0.975	0.814	Best F1; interpretable coefficients
XGBoost (100 rounds)	0.685	0.973	0.804	Best probability calibration.

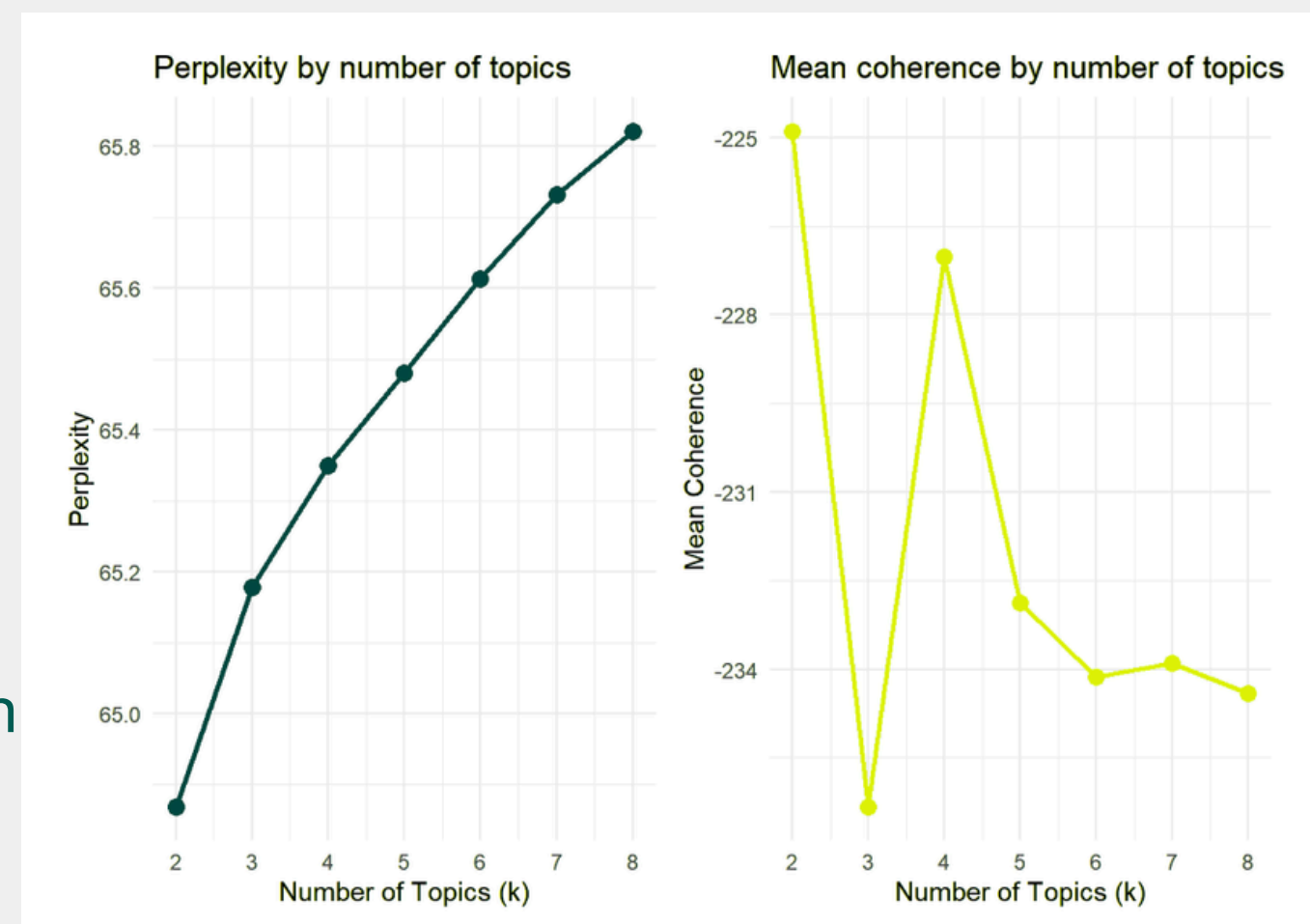
All Models Summary

Error Analysis – Top 5 Confident Mistakes (all false positives)

- Pattern 1: Tweets mentioning complaint vocabulary in a neutral/resolved context (e.g. thanking airline after issue handled)
- Pattern 2: Tweets directed at multiple airlines, complaint context bleeds across mentions
- Root cause: TF-IDF weights rare complaint words heavily; mere presence pushes the model toward negative even without a negative tone.
- Implication: Model would benefit from sentence-level context (motivation for LLM approach in Part 4)

LDA DTM Preparation & Choosing k

- Used raw BOW counts (not TF-IDF). LDA probability model assumes integer counts.
- Full dataset used (not just training split) a total of 14124 documents
- Tokenization identical to Part 1: letters-only, length > 2, stop words removed.
- `removeSparseTerms(sparse = 0.99)` trims vocabulary from 8,926 → 90 terms; 2,737 docs with zero term counts removed before LDA → effective corpus: 11,387 documents



14,124 × 8,926

DTM Before Trimming

14,124 × 90

DTM After Trimming

99%

Sparsity (post-trim)

k selection: Perplexity increases monotonically with k (uninformative on 90-term vocab); coherence peaks at k = 2 but requires extra justification; k = 3 scores lower than k = 2 and k = 4 on coherence (~-235) but is the smallest value beyond k = 2 with operationally meaningful topic separation k = 3 chosen

LDA Topic Interpretation ($k = 3$)

T1: Customer Service & Gate Operations

Key words: flight, service, plane, gate, hours, call, late, change, customer, response
| $n = 3,816$

T2: Flight Cancellations & Booking Disruptions

Key words: flight, cancelled, flights, fly, check, hold, hour, customer, airline, told | $n = 3,972$

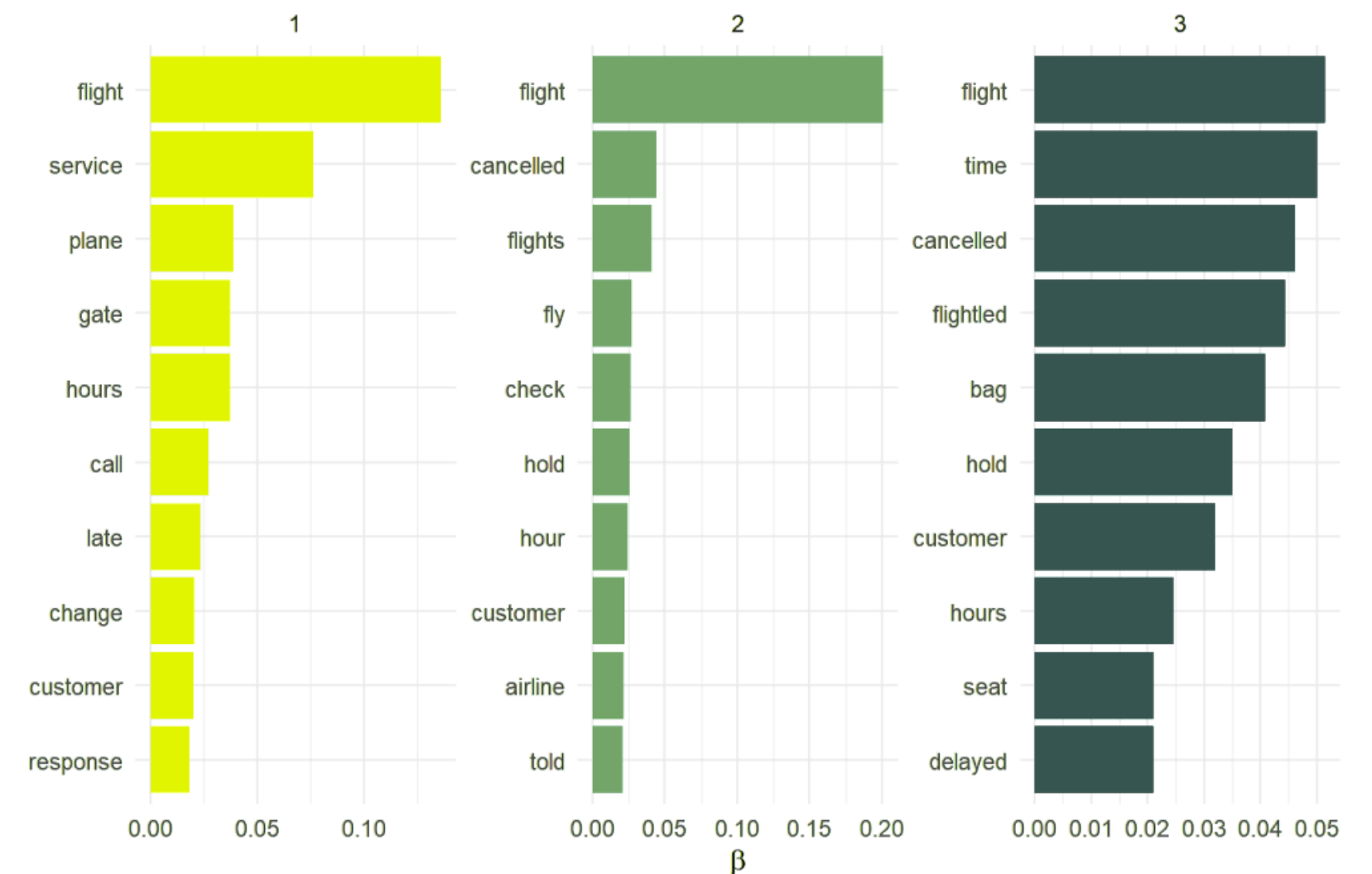
T3: Baggage, Seating & Wait-Time Complaints

Key words: flight, time, cancelled, bag, hold, customer, hours, seat, delayed | $n = 3,599$

LDA Corpus Note

All 3 topics ~65-73% negative. After sparse=0.99 trimming, 2,737 empty docs removed; LDA fitted on 11,387 documents (3,816 + 3,972 + 3,599)

Top 10 words per topic - Twitter Airline Sentiment





Topics vs. Sentiment Distribution

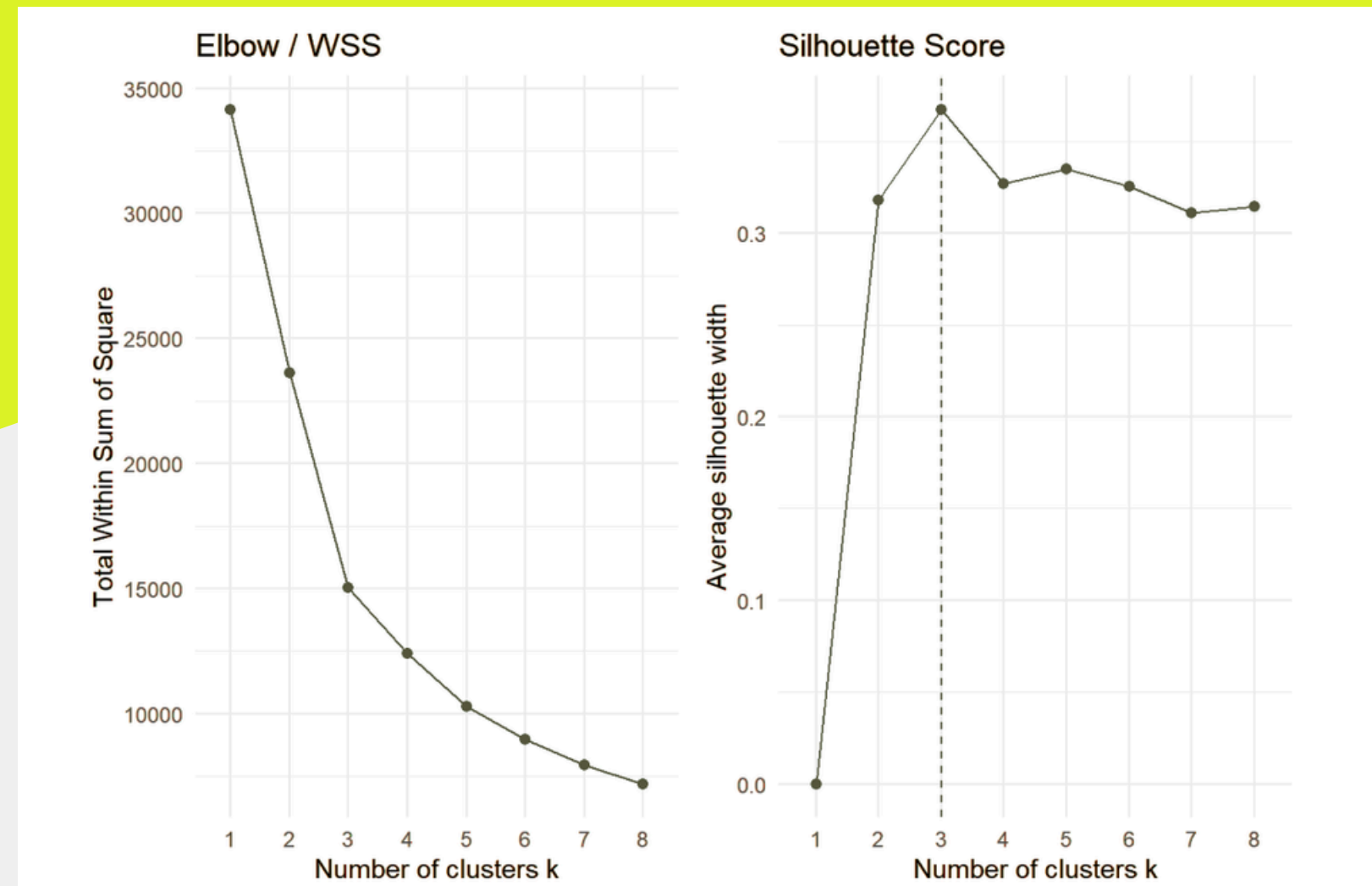


Key insights

- All three topics are 65–73% negative
- Dataset is complaint-driven across every theme
- Topic 1 (Customer Service & Gate) least skewed, some resolved interactions
- Topics 2-3 (Cancellations, Baggage) ~70-73% negative, anger dominates
- LDA reveals the 'why' behind the classifier's binary negative flag
- No single 'praise topic', corpus is structurally complaint-heavy



Customer Segmentation via Clusterin



C1

Baggage, Seat & Wait-Time Complainers (n = 3,475)

Elevated γ on T1 — phone/hold interactions

C2

Cancellation & Booking Complainers (n = 4,402)

Elevated γ on T2/T3 — ops failures, rebooking

C3

Customer Service & Gate Operations Complainers (n = 3,510)

Mixed themes — broad dissatisfaction

Prompt Engineering Experiment

Prompt 1 — Zero-Shot

Accuracy: 19/20 (95%)

Errors: 1 false positive

Misclassified neutral JetBlue seat observation

Prompt 2 — Few-Shot + Sarcasm Rules

Accuracy: 20/20 (100%)

Errors: 0 errors

5 examples + explicit sarcasm handling resolved ambiguous cases

```
# A tibble: 20 × 5
  doc_id text                                true_label Prompt1_pred Prompt2_pred
  <int> <chr>                                <dbl>      <dbl>      <dbl>
1  11563 @US Airways after last night in C...      1          1          1
2   7061 @JetBlue window exits give it aw...      0          1          0
3   5530 @SouthwestAir OMG OMG OMG !!! I ...      0          0          0
4  11806 @US Airways - I had exceptional s...      0          0          0
5   9018 @US Airways I want to love you! 4...      1          1          1
6   3322 @united thanks for making sure t...      0          0          0
7    799 @united well, considering every ...      1          1          1
8   4020 @united Hours on the phone...hun...      1          1          1
9   1480 @united Otis in the baggage cla...      1          1          1
10  12608 @AmericanAir #2298. Everyone els...      1          1          1
11   898 @united how is Mexico not intern...      0          0          0
12   485 @VirginAmerica Grouping Virgin i...      1          1          1
13  12285 @AmericanAir now we are waiting ...      1          1          1
14   9477 @US Airways I called the number a...      1          1          1
15  10594 @usairways I've been on hold for...      1          1          1
16  12508 @AmericanAir another generic res...      1          1          1
17   5554 @SouthwestAir please acknowledge...      0          0          0
18   4653 @SouthwestAir but when do I get ...      1          1          1
19  12756 @AmericanAir ok makes no sense t...      1          1          1
20   1464 @united if you don't like delaye...      1          1          1
```

Key lesson: Few-shot examples + explicit sarcasm/edge-case rules are critical for short, context-dependent tweets. Even 5 examples eliminated the one zero-shot error.

CONCLUSION

Model Bias

- Most top predictors are unambiguously sentiment-bearing (cancelled, delayed, rude vs. kudos, love)
- Airline-brand names may encode carrier-specific bias rather than generalizable negativity
- Short tweets (mobile users) over-predicted as majority class may systematically miss certain user groups

Hallucination Risk

- Zero-shot LLM misclassified '@JetBlue window exits give it away' as negative (neutral seat observation)
- Dangerous scenario: automated response scripts triggered by misclassification sends wrong message to passenger
- Mitigation: human-in-the-loop review before any automated action

Deployment Ethics

- False negatives: missed complaints harm customers; systematic misses for sarcasm users = discriminatory service
- False positives: unnecessary agent workload; intrusive outreach to non-complaining passengers

Recommendation

- Recommended safeguard: confidence-threshold routing → human review (not fully automated responses)

Thank You...